

10TH MATHS QUADRATIC GRAPH SOLUTION

QUADRATIC GRAPH EM NEW (2025 - 2026)

Example: 3.51 Discuss the nature of solutions of the following quadratic equations.

(i) $x^2 + x - 12 = 0$ (ii) $x^2 - 8x + 16 = 0$ (iii) $x^2 + 2x + 5 = 0$

Solution:

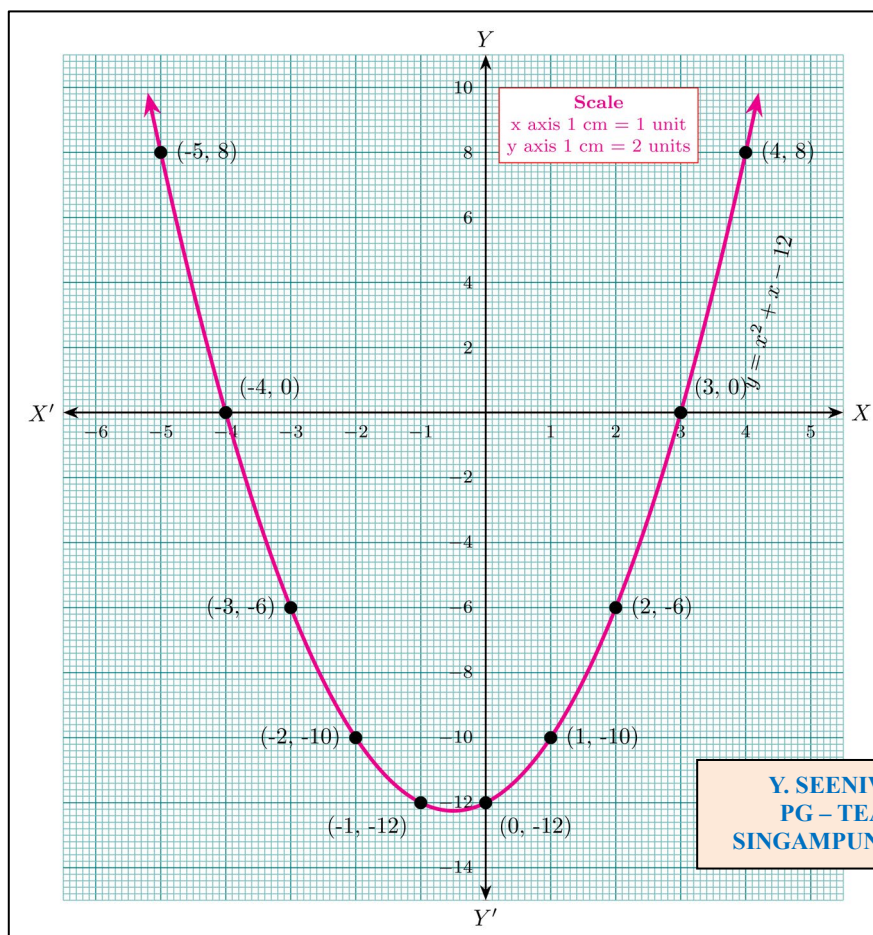
(i) $y = x^2 + x - 12 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-1}{2} = -0.5$ (between -1 to 0 take left 3 point right 3 point)

x	-4	-3	-2	-1	0	1	2	3
x^2	16	9	4	1	0	1	4	9
$+x$	-4	-3	-2	-1	0	1	2	3
-12	-12	-12	-12	-12	-12	-12	-12	-12
y	0	-6	-10	-12	-12	-10	-6	0

Plot the points : (-4, 0), (-3, -6), (-2, -10), (-1, -12), (0, -12), (1, -10), (2, -6), (-4, 0), (3, 0).

$\therefore$ The Quadratic Equation $x^2 + x - 12 = 0$ has **Real and Unequal Roots.**



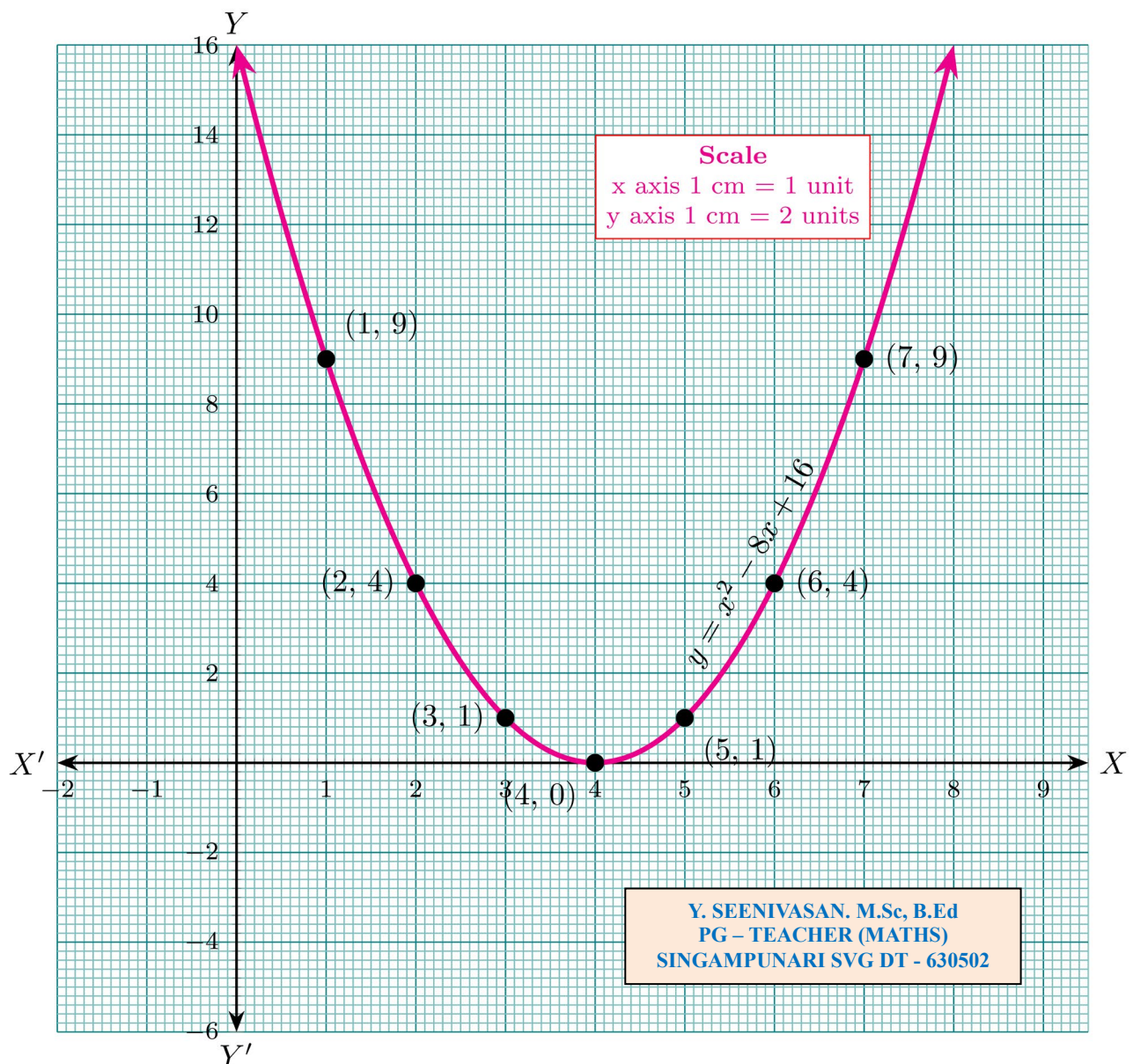
$$(ii) y = x^2 - 8x + 16 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-(-8)}{2} = 4 \text{ (between 4 Left 3 Point and Right 3 point)}$$

x	1	2	3	4	5	6	7
x^2	1	4	9	16	25	36	49
$-8x$	-8	-16	-24	-32	-40	-48	-56
$+16$	16	16	16	16	16	16	16
y	9	4	1	0	1	4	9

Plot the points : (1, 9), (2, 4), (3, 1), (4, 0), (5, 1), (6, 4), (7, 9).

$\therefore$ The Quadratic Equation $x^2 - 8x + 16 = 0$ has **Real and Equal Roots.**



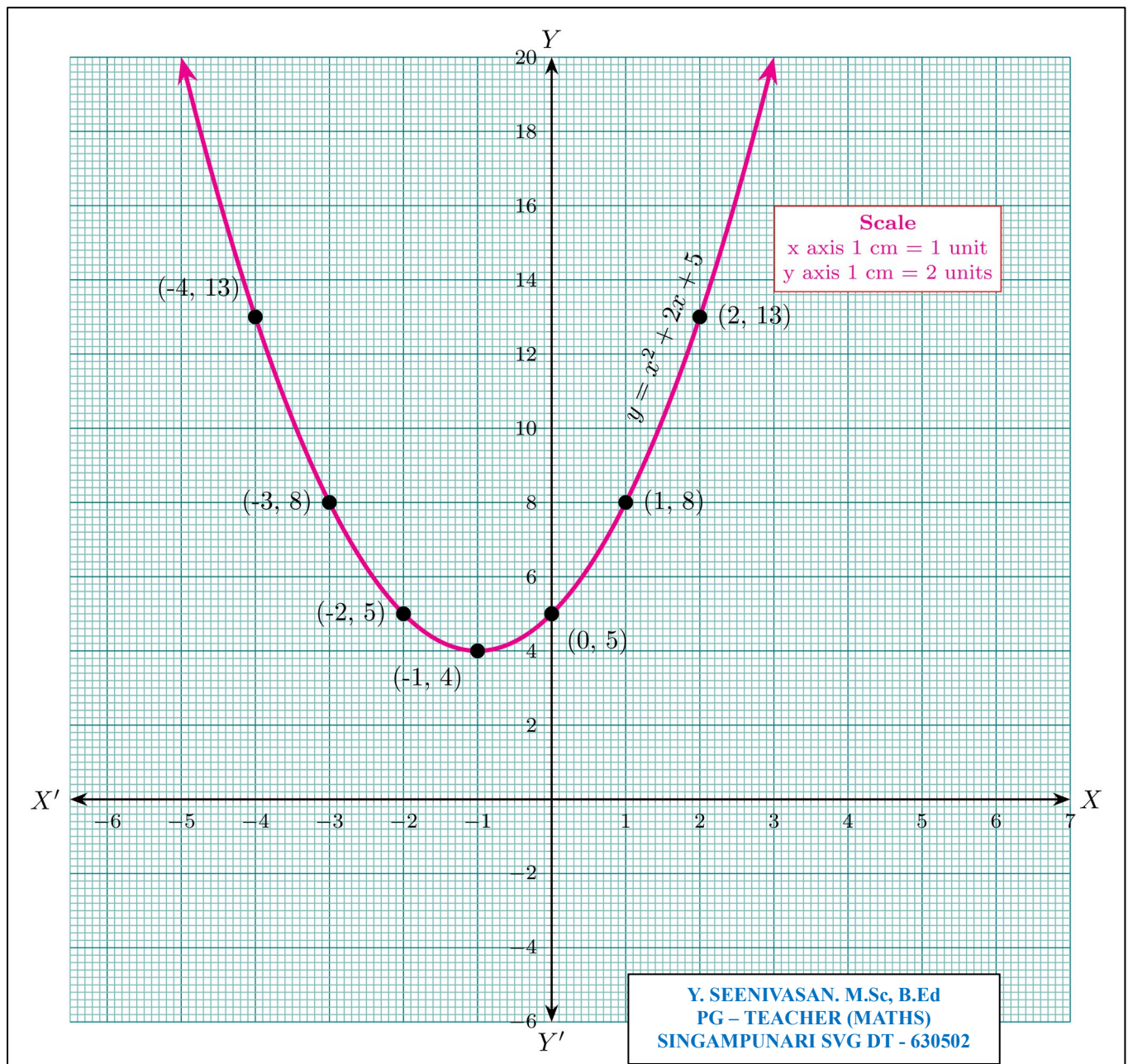
$$(iii) \quad y = x^2 + 2x + 5 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-2}{2} = -1 \text{ (between -1 Left 3 Point and Right 3 point)}$$

x	-4	-3	-2	-1	0	1	2
x^2	16	9	4	1	0	1	4
$+2x$	-8	-6	-4	-2	0	2	4
$+5$	5	5	5	5	5	5	5
y	13	8	5	4	5	8	13

Plot the points : (-4, 13), (-3, 8), (-2, 5), (-1, 4), (0, 5), (1, 8), (2, 13).

$\therefore$ The Quadratic Equation $x^2 - 8x + 16 = 0$ has **No Real Roots.**



Example: 3.52 Draw the graph of $y = 2x^2$ and hence solve $2x^2 - x - 6 = 0$.

Solution:

Given, $y = 2x^2 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{0}{2} = 0$ (between 0 Left 3 Point and Right 3 point)

x	-3	-2	-1	0	1	2	3
x^2	9	4	1	0	1	4	9
$Y = 2x^2$	18	8	2	0	2	8	18

Plot the points : (-3, 18), (-2, 8), (-1, 2), (0,0), (1, 2), (2, 8), (3, 18)

Solve :

$$y = 2x^2$$

$$0 = 2x^2 - x - 6$$

(-) (+) (+)

$$y = x + 6$$

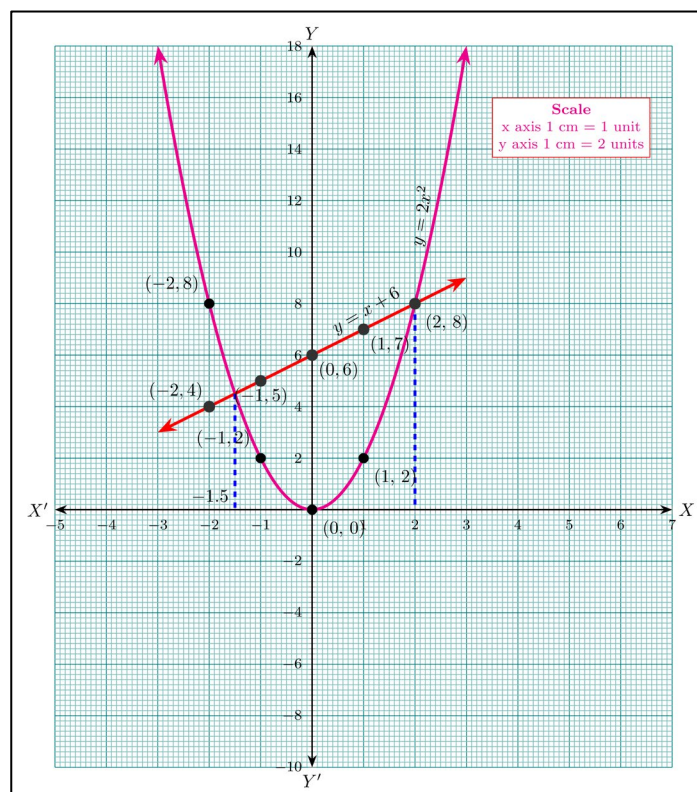
$y = x + 6$

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x	-3	-2	-1	0	1	2	3
$+6$	6	6	6	6	6	6	6
$Y = x + 6$	3	4	5	6	7	8	9

Plot the Points : (-3, 3), (-2, 4), (-1, 5), (0,6), (1, 7), (2, 8), (3, 9)

∴ The solution set of Equation $2x^2 - x - 6 = 0$ has $\{-1.5, 2\}$.



Example: 3.53 Draw the graph of $y = x^2 + 4x + 3$ and hence solve $x^2 + x + 1 = 0$.

Solution:

Given, $y = x^2 + 4x + 3 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-4}{2} = -2$ (between -2 Left 3 Point and Right 3 point)

x	-5	-4	-3	-2	-1	0	1
x^2	25	16	9	4	1	0	1
$+ 4x$	-20	-16	-12	-8	-4	0	4
$+ 3$	+ 3	+ 3	+ 3	+ 3	+ 3	+ 3	+ 3
y	8	3	0	-1	0	3	8

Plot the points : (-5, 8), (-4, 3), (-3, 0), (-2, -1), (-1, 0), (0, 3), (1, 8)

Solve :

$$y = x^2 + 4x + 3$$

$$0 = x^2 + x + 1$$

$$(-) \quad (-) \quad (-)$$

$$y = 3x + 2$$

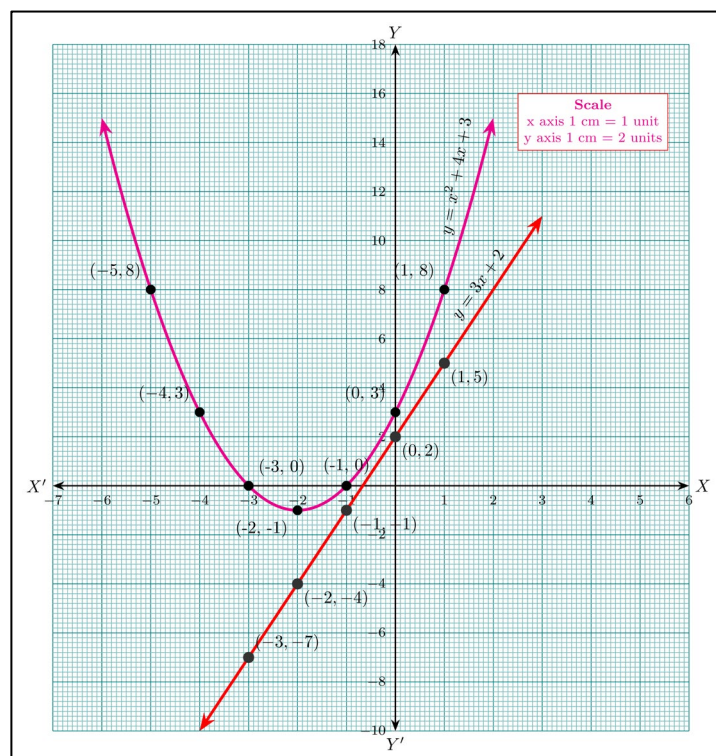
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$$y = 3x + 2$$

x	-4	-3	-2	-1	0	1
$3x$	-12	-9	-6	-3	0	3
$+ 2$	+ 2	+ 2	+ 2	+ 2	+ 2	+ 2
y	-10	-7	-4	-1	2	5

Plot the Points : (-4, -10), (-3, -7), (-2, -4), (-1, -1), (0, 2), (1, 5).

$\therefore$ The solution set of Equation $x^2 + x + 1 = 0$. has **No Real Roots.**



Example: 3.54 Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.

Solution:

$$\text{Given, } y = x^2 + x - 2 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-1}{2} = -0.5 \text{ (between -1 to 0 Left 3 Point and Right 3 point)}$$

x	-4	-3	-2	-1	0	1	2	3
x^2	16	9	4	1	0	1	4	9
$+x$	-4	-3	-2	-1	0	1	2	3
-2	-2	-2	-2	-2	-2	-2	-2	-2
y	10	4	0	-2	-2	0	4	10

Plot the points : $(-4, 10), (-3, 4), (-2, 0), (-1, -2), (0, -2), (1, 0), (2, 4), (3, 10)$

Solve :

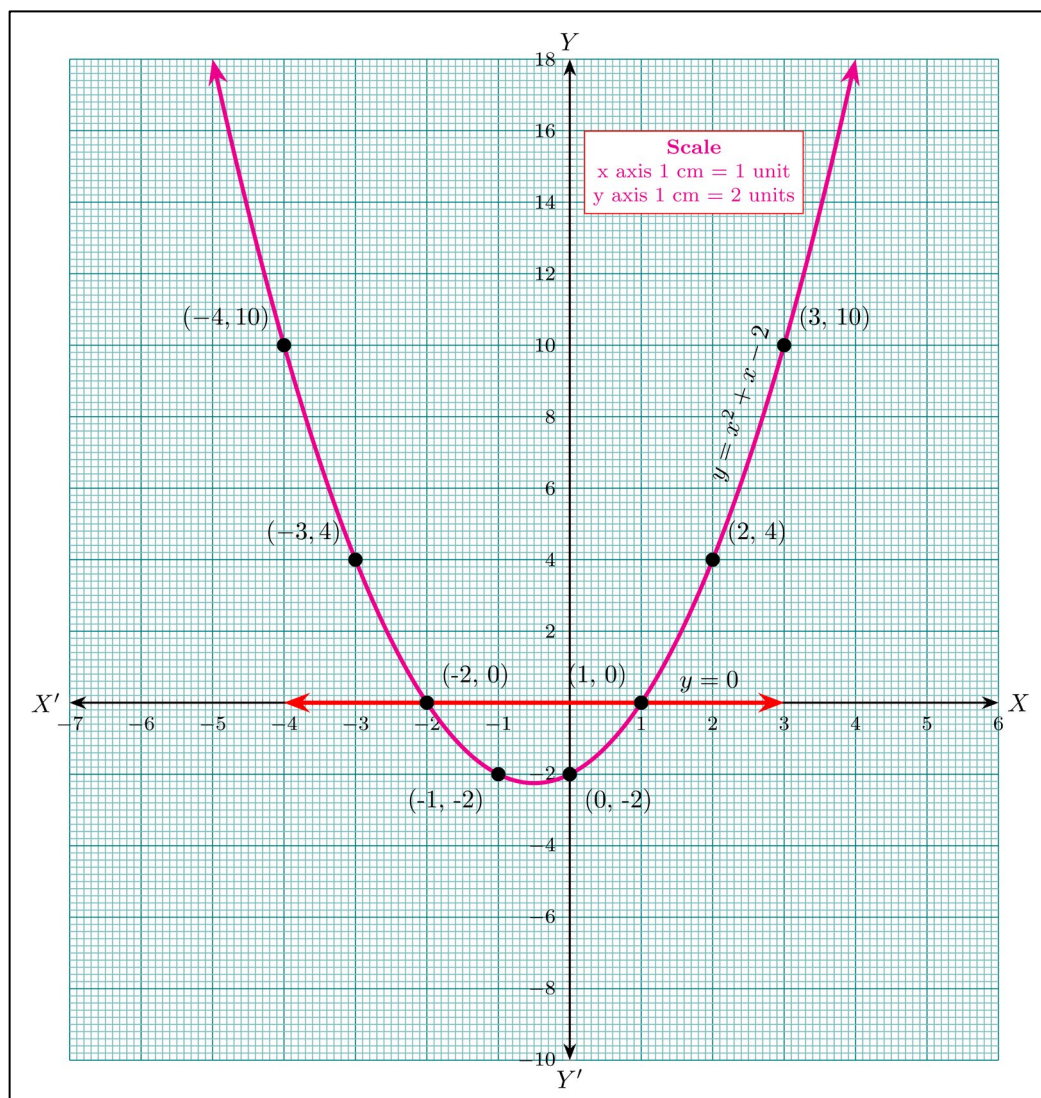
$$y = x^2 + x - 2$$

$$0 = x^2 + x - 2$$

(-) (-) (+)

$$y = 0$$

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Example: 3.55: Draw the graph of $y = x^2 - 4x + 3$ and hence solve $x^2 - 6x + 9 = 0$.

Solution:

Given, $y = x^2 - 4x + 3 \Rightarrow ax^2 + bx + c = 0$

$$\frac{-b}{2a} = \frac{-(-4)}{2} = 2 \text{ (between 2 Left 3 Point and Right 3 point)}$$

x	-1	0	1	2	3	4	5
x^2	1	0	1	4	9	16	25
$-4x$	4	0	-4	-8	-12	-16	-20
$+3$	+3	+3	+3	+3	+3	+3	+3
y	8	3	0	-1	0	3	8

Plot the points : (-1, 8), (0, 3), (1, 0), (2, -1), (3, 0), (4, 3), (5, 8)

Solve :

$$y = x^2 - 4x + 3$$

$$0 = x^2 - 6x + 9$$

(-) (+) (-)

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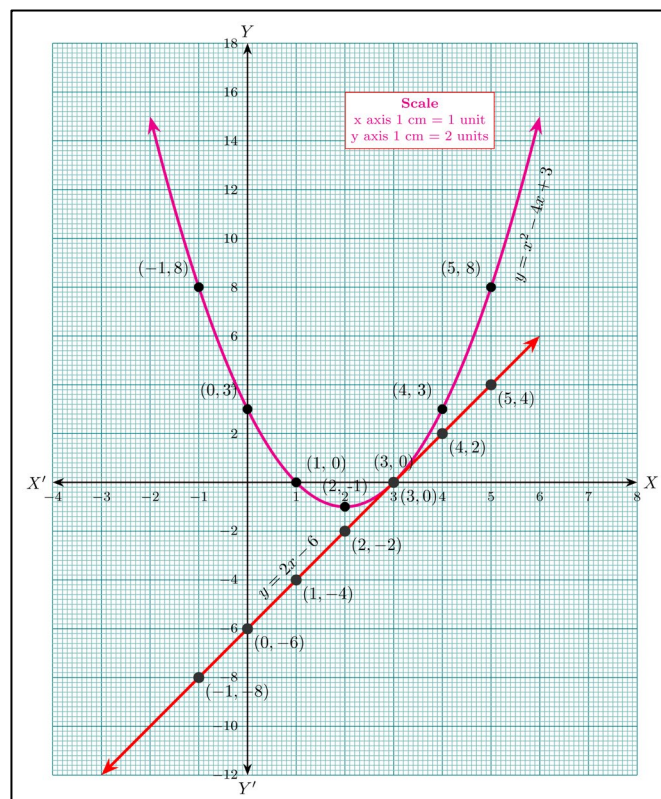
$$y = 2x - 6$$

$$y = 2x - 6$$

x	-1	0	1	2	3	4
$2x$	-2	0	2	4	6	8
-6	-6	-6	-6	-6	-6	-6
y	-8	-6	-4	-2	0	2

Plot the Points : (-1, -8), (0, -6), (1, -4), (2, -2), (3, 0), (4, 2).

$\therefore$ The solution set of Equation $x^2 - 6x + 9 = 0$ has {3} only .



Exercise: 3.16 1) Graph the following quadratic equations and state the nature of solutions.

- (i) $x^2 - 9x + 20 = 0$ (ii) $x^2 - 4x + 4 = 0$ (iii) $x^2 + x + 7 = 0$ (iv) $x^2 - 9 = 0$
 (v) $x^2 - 6x + 9 = 0$ (vi) $(2x - 3)(x + 2) = 0$

Solution:

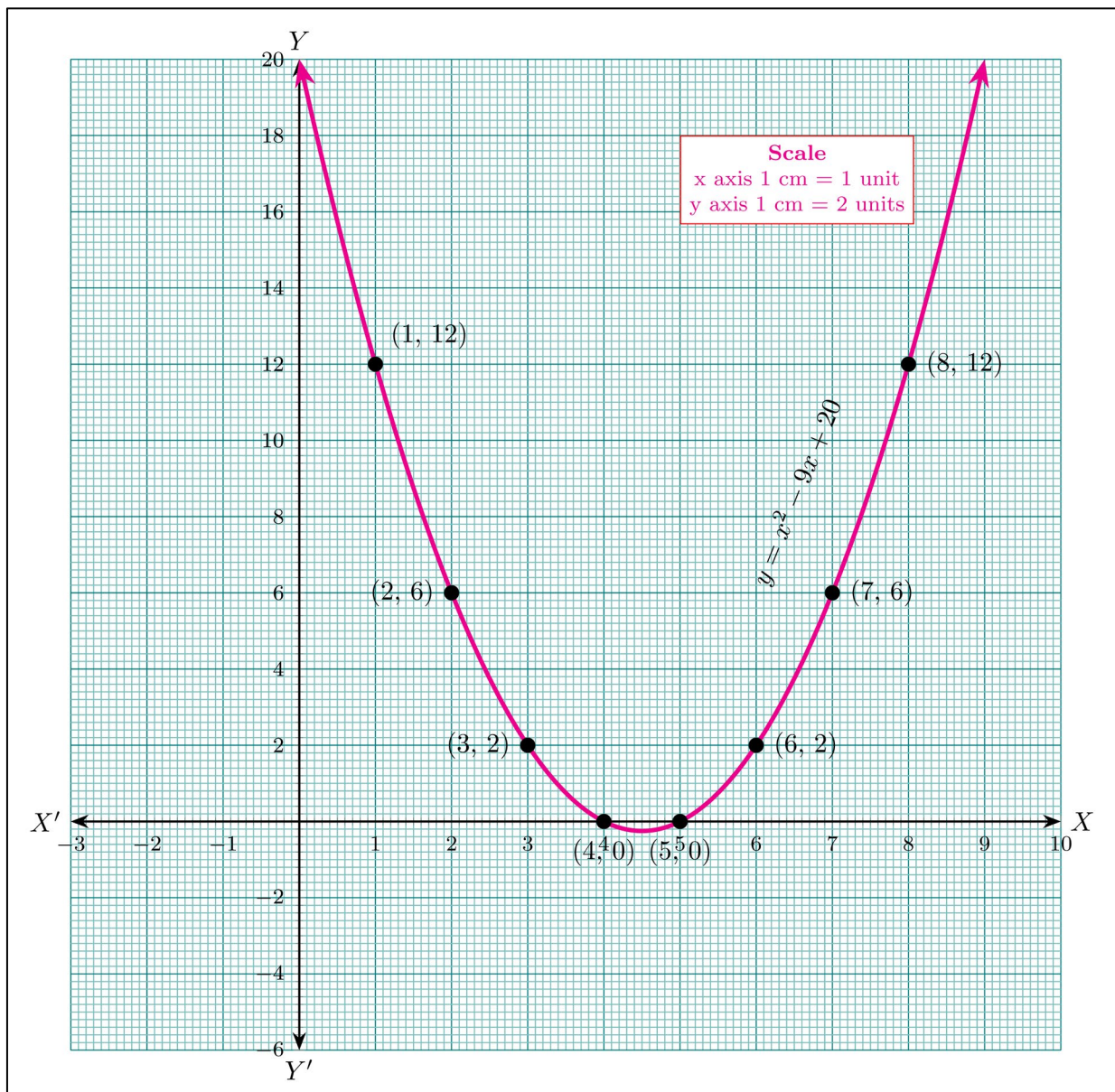
$$(i) y = x^2 - 9x + 20 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-(-9)}{2} = 4.5 \text{ (between 4 to 5 take left 3 point right 3 point)}$$

x	1	2	3	4	5	6	7	8
x^2	1	4	9	16	25	36	49	64
$-9x$	-9	-18	-27	-36	-45	-54	-63	-72
$+20$	+20	+20	+20	+20	+20	+20	+20	+20
y	12	6	2	0	0	2	6	12

Plot the points : (1, 12), (2, 6), (3, 2), (4, 0), (5, 0), (6, 2), (7, 6), (8, 12).

$\therefore$ The Quadratic Equation $x^2 - 9x + 20 = 0$ has Real and Unequal Roots.



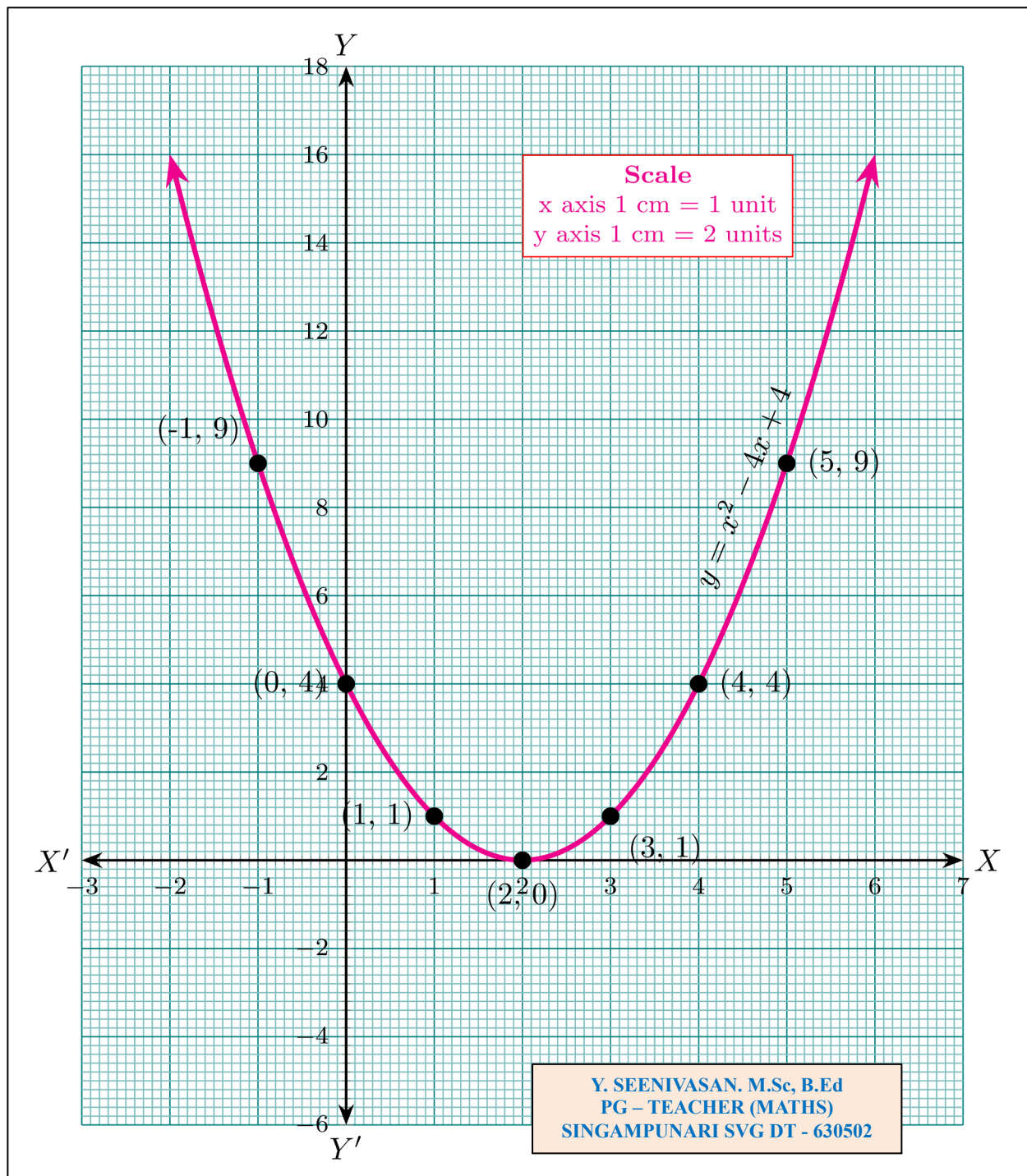
$$(ii) y = x^2 - 4x + 4 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-(-4)}{2} = 2 \text{ (between 2 take left 3 point right 3 point)}$$

x	-1	0	1	2	3	4	5
x^2	1	0	1	4	9	16	25
$-4x$	4	0	-4	-8	-12	-16	-20
$+4$	$+4$	$+4$	$+4$	$+4$	$+4$	$+4$	$+4$
y	9	4	1	0	1	4	9

Plot the points : (-1, 9), (0, 4), (1, 1), (2, 0), (3, 1), (4, 4), (5, 9).

$\therefore$ The Quadratic Equation $x^2 - 4x + 4 = 0$ has Real and Equal Roots.



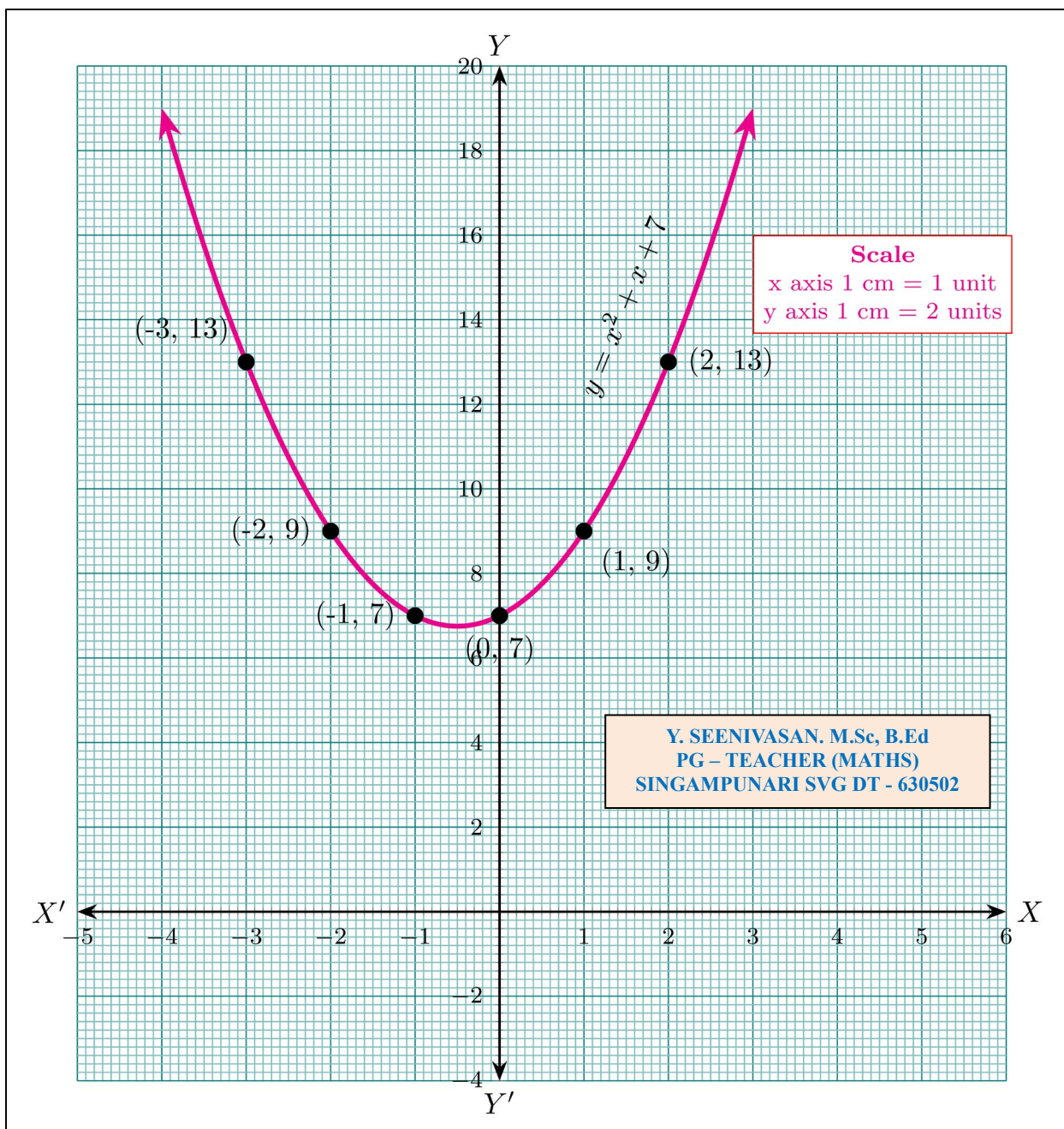
$$(iii) y = x^2 + x + 7 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-1}{2} = -0.5 \text{ (between -1 to 0 take left 3 point right 3 point)}$$

x	-4	-3	-2	-1	0	1	2	3
x^2	16	9	4	1	0	1	4	9
$+x$	-4	-3	-2	-1	0	1	2	3
$+7$	+7	+7	+7	+7	+7	+7	+7	+7
y	19	13	9	7	7	9	13	19

Plot the points : (-4, 19), (-3, 13), (-2, 9), (-1, 7), (0, 7), (1, 9), (2, 13), (3, 19).

$\therefore$ The Quadratic Equation $x^2 + x + 7 = 0$ has No Real Roots.



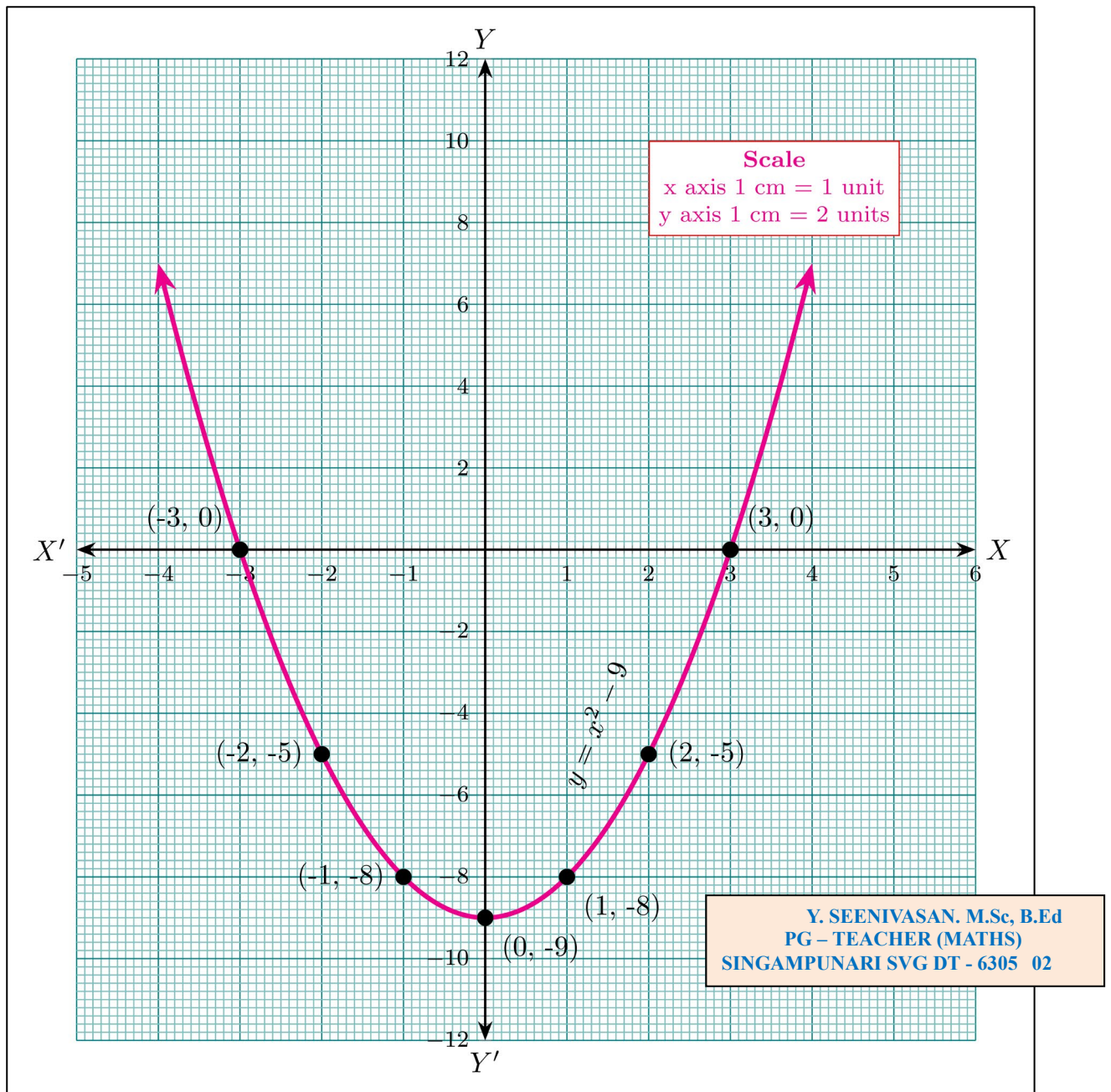
$$(iv) y = x^2 - 9 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{0}{2} = 0 \text{ (between 0 take left 3 point right 3 point)}$$

x	-3	-2	-1	0	1	2	3
x^2	9	4	1	0	1	4	9
-9	-9	-9	-9	-9	-9	-9	-9
y	0	-5	-8	-9	-8	-5	0

Plot the points : (-3, 0), (-2, -5), (-1, -8), (0, -9), (1, -8), (2, -5), (3, 0).

$\therefore$ The Quadratic Equation $x^2 - 9 = 0$ has Real and Unequal Roots.



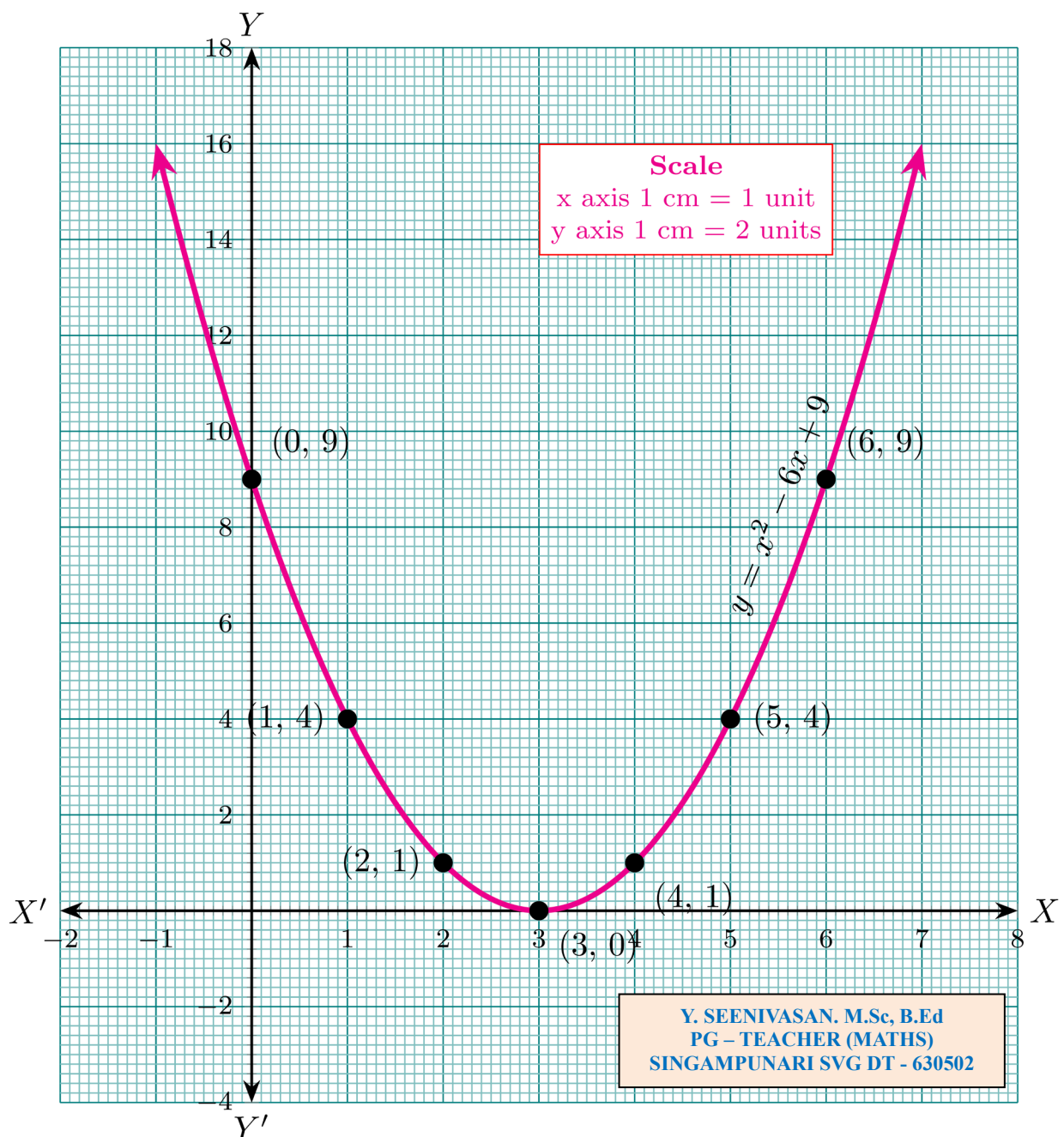
$$(v) y = x^2 - 6x + 9 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-(-6)}{2} = 3 \text{ (between 3 take left 3 point right 3 point)}$$

x	0	1	2	3	4	5	6
x^2	0	1	4	9	16	25	36
$-6x$	0	-6	-12	-18	-24	-30	-36
$+9$	+9	+9	+9	+9	+9	+9	+9
y	9	4	1	0	1	4	9

Plot the points : (0, 9), (1, 4), (2, 1), (3, 0), (4, 1), (5, 4), (6, 9).

$\therefore$ The Quadratic Equation $x^2 - 6x + 9 = 0$ has Real and Equal Roots.



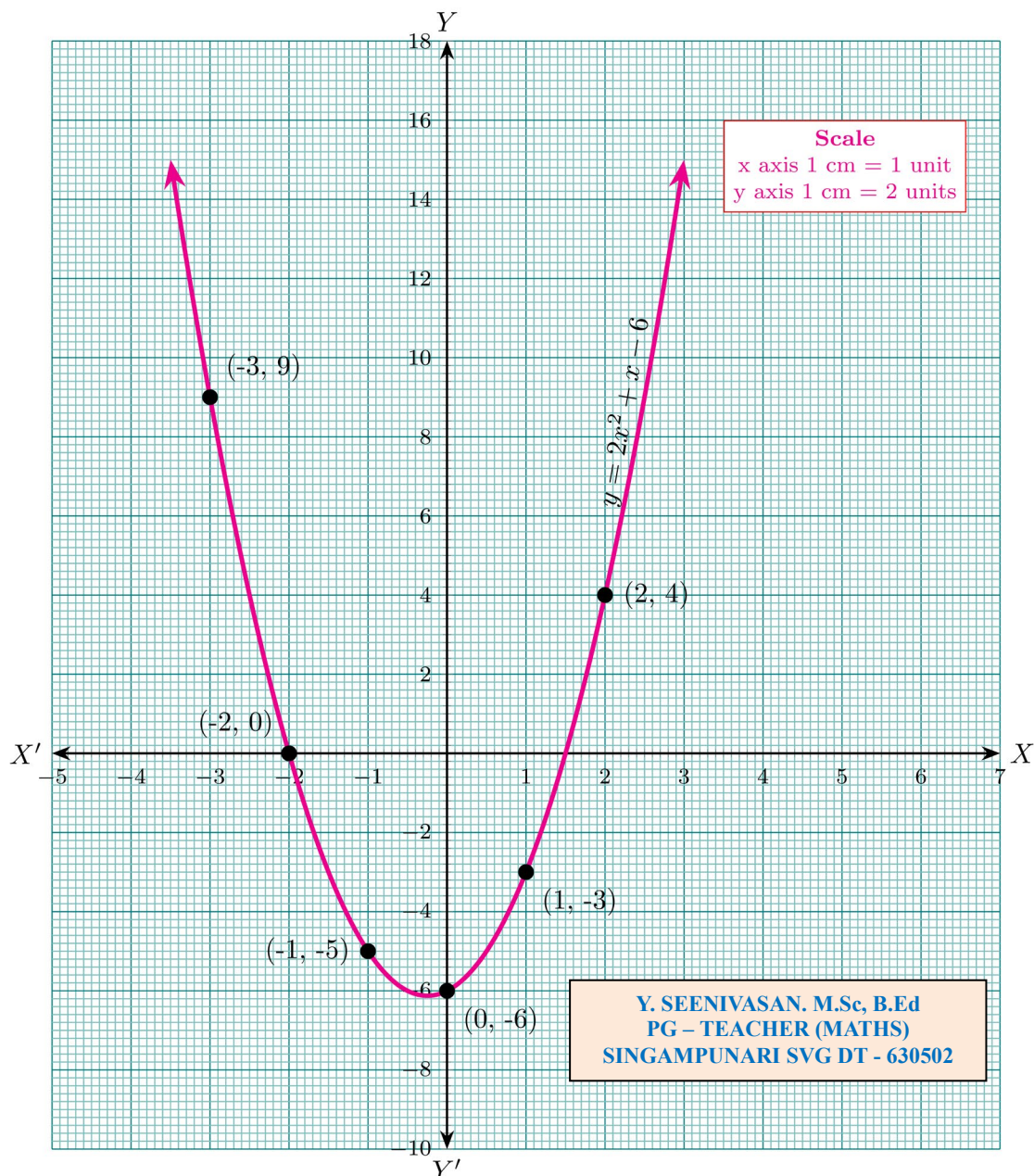
$$(vi) y = (2x - 3)(x + 2) = 2x^2 + 4x - 3x - 6 = 2x^2 + x - 6 = 0 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-1}{2 \times 2} = \frac{-1}{4} = -0.25 \text{ (between -1 to 0 take left 3 point right 3 point).}$$

x	-4	-3	-2	-1	0	1	2	3
$2x^2$	32	18	8	2	0	2	8	18
$+x$	-4	-3	-2	-1	0	1	2	3
-6	-6	-6	-6	-6	-6	-6	-6	-6
y	22	9	0	-5	-6	-3	4	15

Plot the points : (-4, 22), (-3, 9), (-2, 0), (-1, -5), (0, -6), (1, -3), (2, 4), (3, 15).

$\therefore$ The Quadratic Equation $(2x - 3)(x + 2) = 0$ has Real and Unequal Roots.



Exercise: 3.16) 2) Draw the graph of $y = x^2 - 4$ and hence solve $x^2 - x - 12 = 0$.

Solution:

$$\text{Given, } y = x^2 - 4 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{0}{2} = 0 \text{ (between 0 Left 3 Point and Right 4 point)}$$

x	-3	-2	-1	0	1	2	3
x^2	9	4	1	0	1	4	9
-4	-4	-4	-4	-4	-4	-4	-4
y	5	0	-3	-4	-3	0	5

Plot the points : (-3, 5), (-2, 0), (-1, -3), (0, -4), (1, -3), (2, 0), (3, 5)

Solve :

$$y = x^2 - 4$$

$$0 = x^2 - x - 12$$

$$(-) \quad (+) \quad (+)$$

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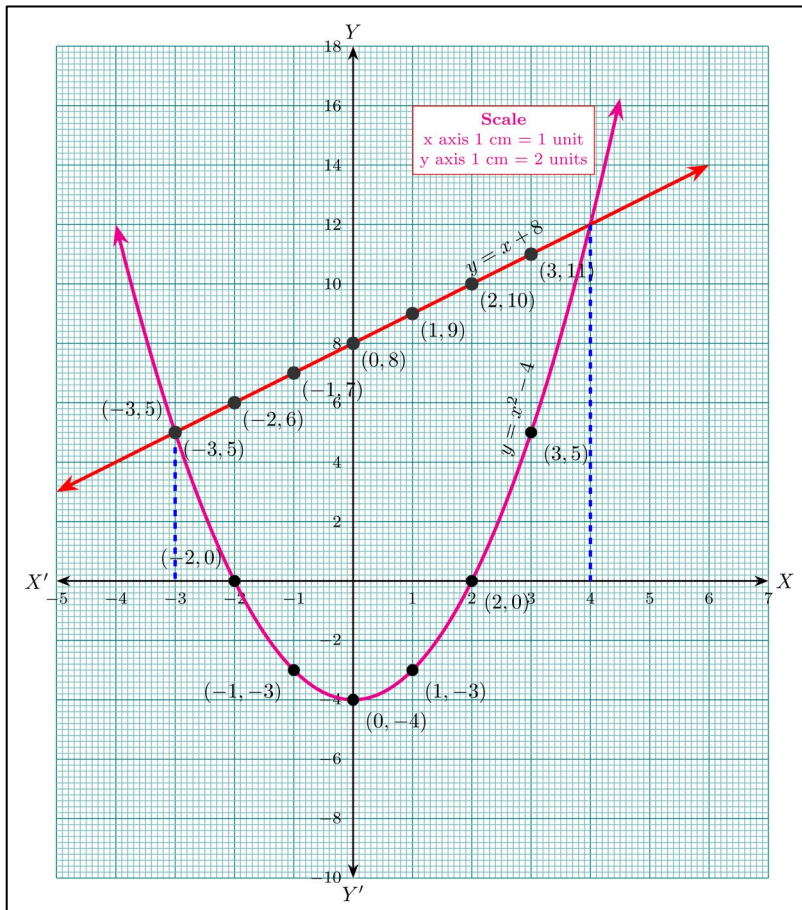
$$y = x + 8$$

$$y = x + 8$$

x	-3	-2	-1	0	1	2	3	4
8	8	8	8	8	8	8	8	8
Y	5	6	7	8	9	10	11	12

Plot the Points : (-3, 5), (-2, 6), (-1, 7), (0, 8), (1, 9), (2, 10), (3, 11).

$\therefore$ The solution set of Equation $x^2 - x - 12 = 0$ has $\{-3, 4\}$.



Exercise: 3.16) 3) Draw the graph of $y = x^2 + x$ and hence solve $x^2 + 1 = 0$.

Solution:

Given, $y = x^2 + x \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-1}{2} = -0.5$ (between -1 to 0 Left 3 Point and Right 3 point)

x	-4	-3	-2	-1	0	1	2	3
x^2	16	9	4	1	0	1	4	9
$+x$	-4	-3	-2	-1	0	1	2	3
y	12	6	2	0	0	2	6	12

Plot the points : (-4, 12), (-3, 6), (-2, 2), (-1, 0), (0, 0), (1, 2), (2, 6), (3, 12).

Solve :

$$y = x^2 + x$$

$$0 = x^2 + 0x + 1$$

(-) (-) (-)

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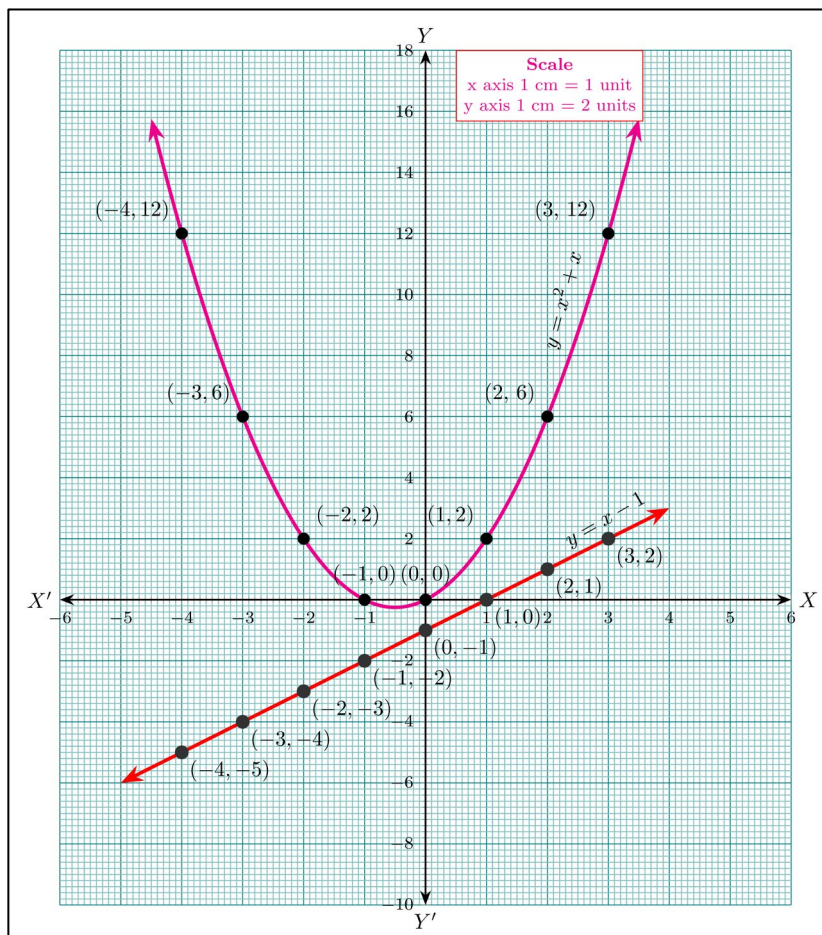
$$y = x - 1$$

$y = x - 1$

x	-4	-3	-2	-1	0	1	2	3
-1	-1	-1	-1	-1	-1	-1	-1	-1
Y	-5	-4	-3	-2	-1	0	1	2

Plot the Points : (-4, -5), (-3, -4), (-2, -3), (-1, -2), (0, -1), (1, 0), (2, 1), (3, 2).

$\therefore$ The solution set of Equation $x^2 + 1 = 0$ has **No Real roots.**



Exercise: 3.16) 4) Draw the graph of $y = x^2 + 3x + 2$ and hence solve $x^2 + 2x + 1 = 0$.

Solution:

$$\text{Given, } y = x^2 + 3x + 2 \Rightarrow ax^2 + bx + c = 0$$

$$\frac{-b}{2a} = \frac{-3}{2} = -1.5 \text{ (between -2 to -1 Left 3 Point and Right 3 point)}$$

x	-5	-4	-3	-2	-1	0	1	2
x^2	25	16	9	4	1	0	1	4
$+3x$	-15	-12	-9	-6	-3	0	3	6
$+2$	+2	+2	+2	+2	+2	+2	+2	+2
y	12	6	2	0	0	2	6	12

Plot the points : $(-5, 12), (-4, 6), (-3, 2), (-2, 0), (-1, 0), (0, 2), (1, 6), (2, 12)$.

Solve :

$$y = x^2 + 3x + 2$$

$$0 = x^2 + 2x + 1$$

$$(-) \quad (-) \quad (-)$$

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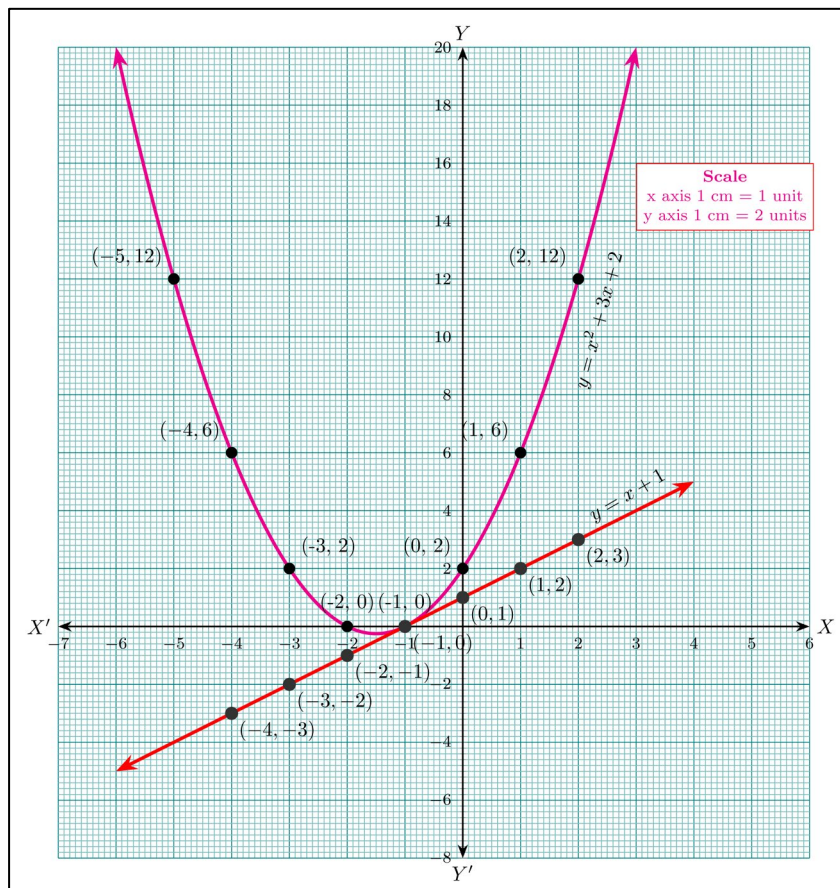
$$y = x + 1$$

$$y = x + 1$$

x	-4	-3	-2	-1	0	1	2
$+1$	+1	+1	+1	+1	+1	+1	+1
Y	-3	-2	-1	0	1	2	3

Plot the Points : $(-4, -3), (-3, -2), (-2, -1), (-1, 0), (0, 1), (1, 2), (2, 3)$.

$\therefore$ The solution set of Equation $x^2 + 2x + 1 = 0$ has $\{-1\}$.



Exercise: 3.16) 5) Draw the graph of $y = x^2 + 3x - 4$ and hence solve $x^2 + 3x - 4 = 0$.

Solution:

Given, $y = x^2 + 3x - 4 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-3}{2} = -1.5$ (between -2 to -1 Left 3 Point and Right 3 point)

x	-5	-4	-3	-2	-1	0	1	2
x^2	25	16	9	4	1	0	1	4
$+3x$	-15	-12	-9	-6	-3	0	3	6
-4	-4	-4	-4	-4	-4	-4	-4	-4
y	6	0	-4	-6	-6	-4	0	6

Plot the points : (-5, 6), (-4, 0), (-3, -4), (-2, -6), (-1, -6), (0, -4), (1, 0), (2, 6).

Solve :

$$y = x^2 + 3x - 4$$

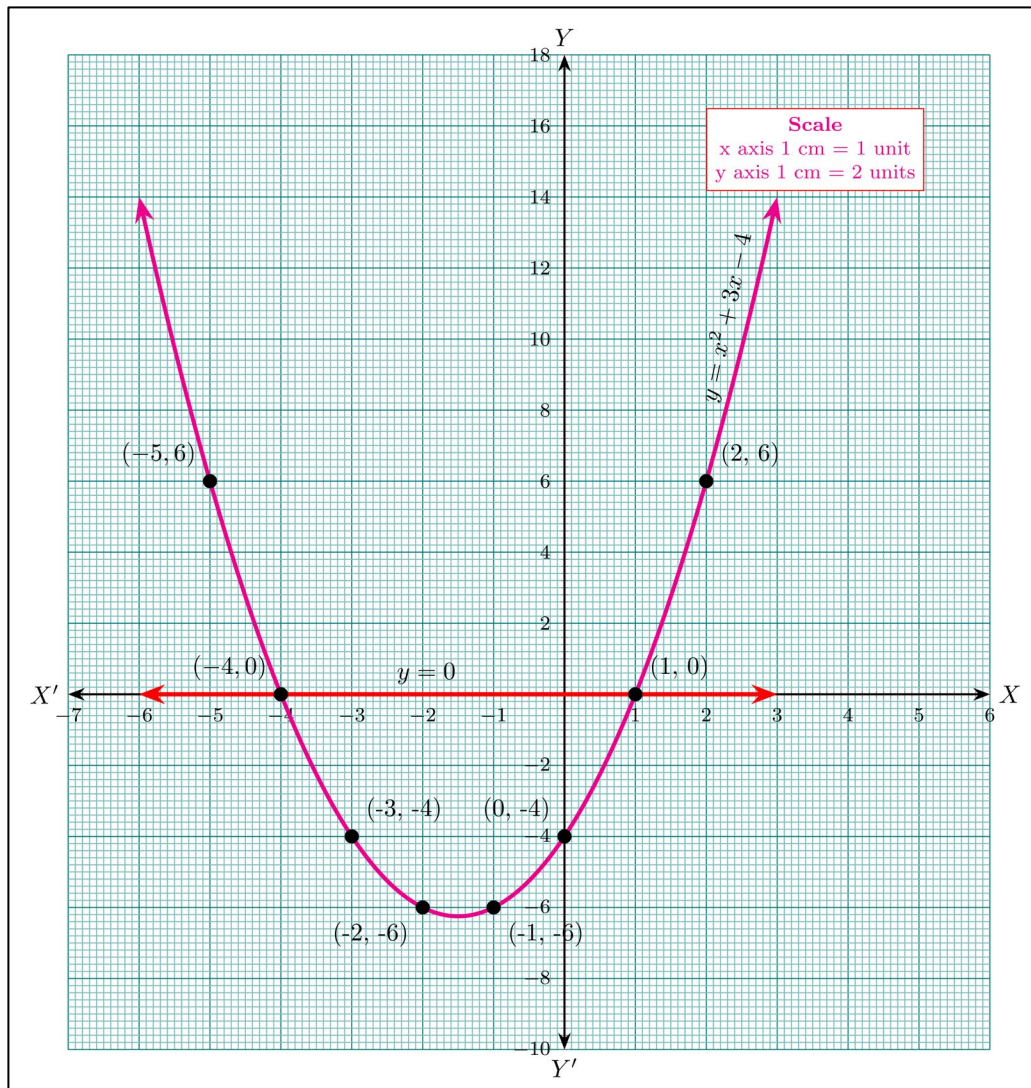
$$0 = x^2 + 3x - 4$$

$$(-) \quad (-) \quad (+)$$

$$y = 0$$

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$\therefore$ The solution set of Equation $x^2 + 3x - 4 = 0$ has $\{-4, 1\}$.



Exercise: 3.16) 6) Draw the graph of $y = x^2 - 5x - 6$ and hence solve $x^2 - 5x - 14 = 0$.

Solution:

Given, $y = x^2 - 5x - 6 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-(-5)}{2} = 2.5$ (between 2 to 3 Left 4 Point and Right 4 point)

x	-2	-1	0	1	2	3	4	5	6	7
x^2	4	1	0	1	4	9	16	25	36	49
$-5x$	10	5	0	-5	-10	-15	-20	-25	-30	-35
-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6
y	8	0	-6	-10	-12	-12	-10	-6	0	8

Plot the points : $(-2, 8), (-1, 0), (0, -6), (1, -10), (2, -12), (3, -12), (4, -10), (5, -6), (6, 0), (7, 8)$.

Solve :

$$y = x^2 - 5x - 6$$

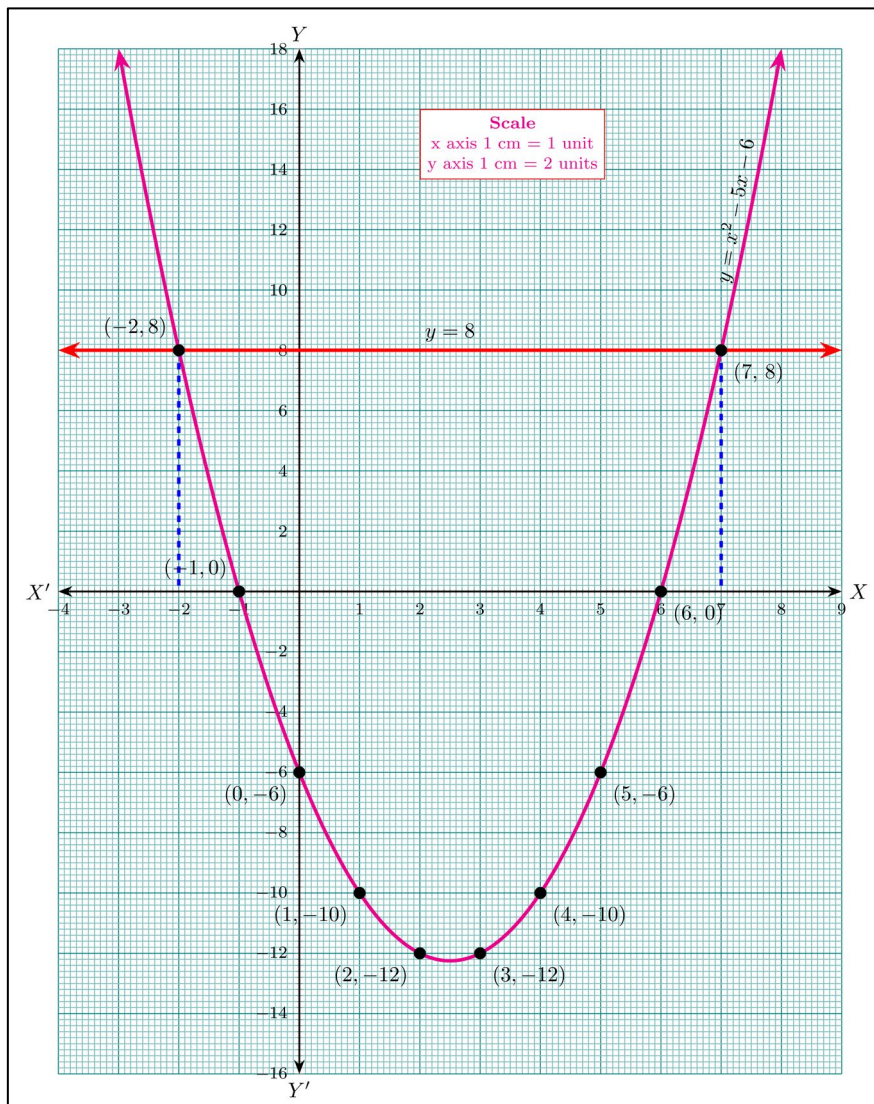
$$0 = x^2 - 5x - 14$$

(-) (+) (+)

$y = 8$

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$\therefore$ The solution set of Equation $x^2 - 5x - 14 = 0$ has $\{-2, 7\}$.



Exercise: 3.16) 7) Draw the graph of $y = 2x^2 - 3x - 5$ and hence solve $2x^2 - 4x - 6 = 0$.

Solution:

Given, $y = 2x^2 - 3x - 5 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-(-3)}{2 \times 2} = \frac{3}{4} = 0.75$ (between 0 to 1 Left 3 Point and Right 3 point)

x	-2	-1	0	1	2	3	4	5
x^2	8	2	0	2	8	18	32	50
$-3x$	6	3	0	-3	-6	-9	-12	-15
-5	-5	-5	-5	-5	-5	-5	-5	-5
y	9	0	-5	-6	-3	4	15	30

Plot the points : (-2, 9), (-1, 0), (0, -5), (1, -6), (2, -3), (3, 4), (4, 15), (5, 30).

Solve :

$$y = 2x^2 - 3x - 5$$

$$0 = 2x^2 - 4x - 6$$

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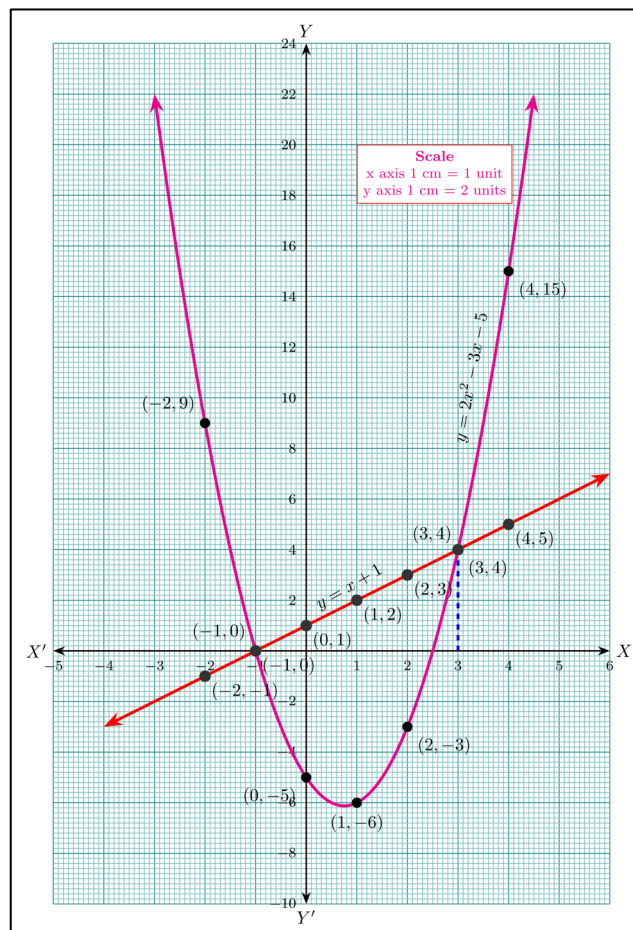
$$y = x + 1$$

$y = x + 1$

x	-2	-1	0	1	2	3	4	5
$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$	$+1$
Y	-1	0	1	2	3	4	5	6

Plot the Points : (-2, -1), (-1, 0), (0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6).

$\therefore$ The solution set of Equation $2x^2 - 4x - 6 = 0$ has $\{-1, 3\}$.



Exercise: 3.16) 8) Draw the graph of $y = (x - 1)(x + 3)$ and hence solve $x^2 - x - 6 = 0$.

Solution:

Given, $y = (x - 1)(x + 3) = x^2 + 3x - x - 3 = x^2 + 2x - 3 \Rightarrow ax^2 + bx + c = 0$

$\frac{-b}{2a} = \frac{-2}{2} = -1$ (between 1 Left 3 Point and Right 4 point)

x	-4	-3	-2	-1	0	1	2
x^2	16	9	4	1	0	1	4
$+2x$	-8	-6	-4	-2	0	2	4
-3	-3	-3	-3	-3	-3	-3	-3
y	5	0	-3	-4	-3	0	5

Plot the points : $(-4, 5), (-3, 0), (-2, -3), (-1, -4), (0, -3), (1, 0), (2, 5)$.

Solve :

$$y = x^2 + 2x - 3$$

$$0 = x^2 - x - 6$$

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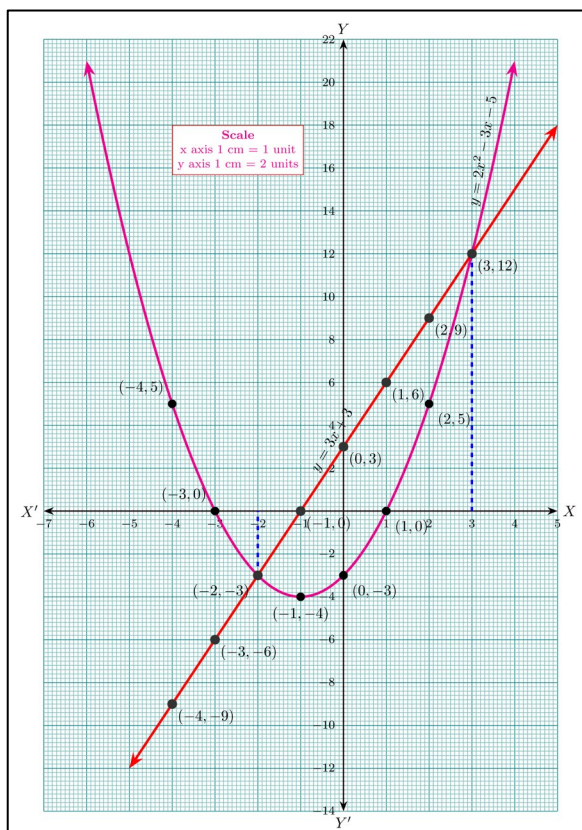
$$y = 3x + 3$$

$$y = 3x + 3$$

x	-4	-3	-2	-1	0	1	2	3
$3x$	-12	-9	-6	-3	0	3	6	9
$+3$	+3	+3	+3	+3	+3	+3	+3	+3
y	-9	-6	-3	0	3	6	9	12

Plot the Points : $(-4, -9), (-3, -6), (-2, -3), (-1, 0), (0, 3), (1, 6), (2, 9), (3, 12)$.

$\therefore$ The solution set of Equation $x^2 - x - 6 = 0$ has $\{-2, 3\}$.



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